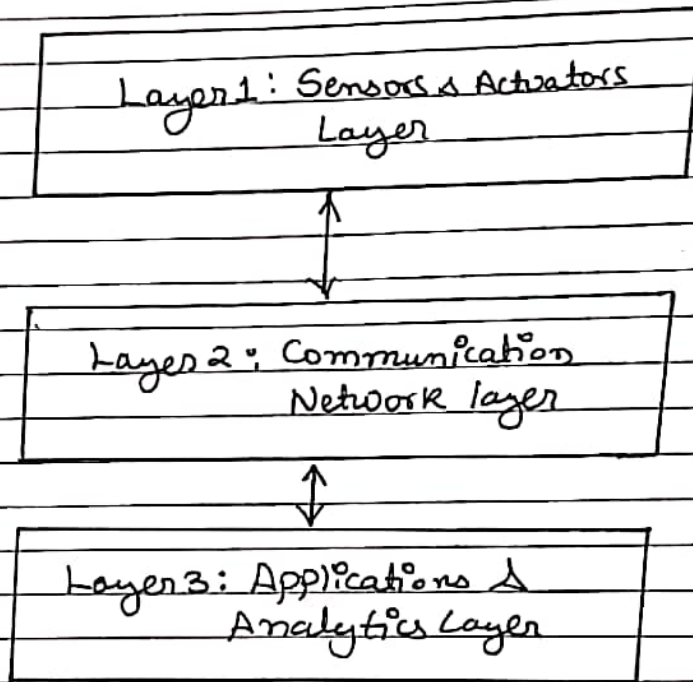


3. Core IoT Functional Stack

#. Introduction

Core IoT Functional Stack comprises of three layers.



- Layer 1 majorly talks about the Sensors and their various types.
- Layer two talks about processing for which microcontrollers are used also it talks about connectivity for which Internet or bluetooth can be used.

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- The third layer talks about data analysis and its presentation. Data incoming from layer 2, layer three processes the data and presents the data in a particular way to the user.

Layer 1: Things: Sensors & Actuators

Q What are the different types of Sensors

- If we aggregate all the sensors from ^{characteristics} Sensor category point of view here are five types of sensors:

1. Battery powered or power connected:
This segregation deals with whether the object carries its own power supply or gets a continuous power from some external power source.

2. Mobile or Static:

This segregation is based on whether the object or thing or sensor shall move or always remain at a particular location.

A sensor can be mobile and the reason might be it is moved from one object to other because it was attached to a moving object, for example a location tracker on the neck belts of pets. At the same time there are sensors which are static as well.

3. Low or High reporting frequency:

This segregation is based on what is the frequency on which an object shall report monitored parameters.

A rust sensor can report values just once every month & on the other hand a motion sensor can report acceleration multiple hundreds of time every second.

4. Simple or rich data:

This Segregation is based on the quantity of data exchanged in every reported cycle.

A humidity sensor in a field can report a simple daily index value and on the other hand an engine sensor can report multiple parameters like gas velocity, compression speed etc.

5. Report range:

This Segregation is based on the distance at which element is located.

For example for Bluetooth headphones to communicate with phone via Bluetooth, it needs to be located a few meters away at max.

Layer Three: Application and Analytics Layer

- The responsibility of this layer is regarding interaction with the end user. The data which is generated by the sensor is responsible to provide some information through probability or prediction and thus it is sent on the cloud for further analysis.

* Analytics Application

Normally it collects data from number of sensors, process the data and represent it according to the end users need in a particular kind of visualisation criteria.

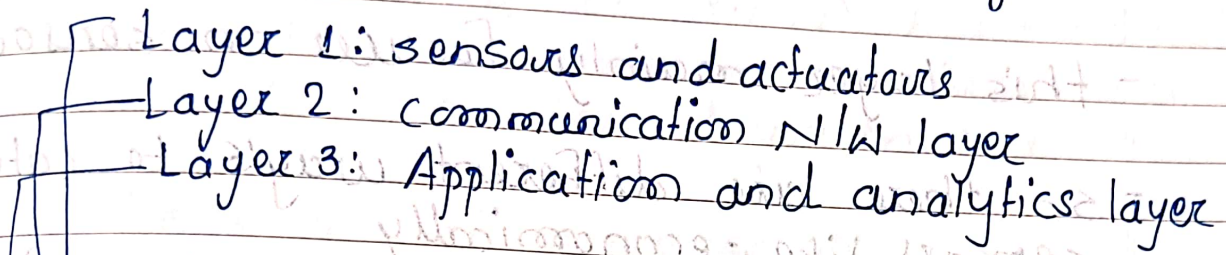
* Control Application

- These type of application ~~and~~ control is the doing or actions of the smart object, or action of an object which is related to the smart object.
- For example a pressure sensor could be connected to a pump. A control application ~~up~~^{up}s the pump speed when the connected sensor detects a drop in pressure.

3. The core IOT Fundamental stack

* Layer 1 - Things: Sensors and Actuators Layer

- You need to know about IOT would forum standardized architecture.
- so they have compressed that layer and made



→ This layer focuses on sensors majorly

→ This layer focuses on processing and connectivity for processing it use micro-controller for connectivity it use blue tooth or internet

→ This layer focuses on data analysis and visualization,

* Layer 1: Things: sensors and actuator layer

- when you want to accept i/p from any IOT device so you must required sensors and at same time you want to present output then actuators are required

- this layer mainly focuses on sensors:

- so there are different ways to categorise sensors like:

- economically
- environmental point of view
- characteristics point of view

→ So if we ~~can~~ ~~consider~~ categorise sensors in characteristics point of view so it comes in five categories:

1) Battery power or power connected sensors

- So these are the sensors where object carries there own power
e.g. battery

2) - or object receives continuous power from external source.

2) Mobile or static:

- so the mobile sensors can move from one location to another. e.g. ~~lights~~ smart cleaner

- static sensors have there fix place
e.g. lights

3) Low or high reporting frequency sensors:
 - these are the sensors in which if you select parameters then the sensing capability of that sensor is either high or low
 - so in high: smart watch because it shows the report of calories burn on daily basis and continue in low: rust sensor in iron rust sensor can report you in every month.

4) Simple and rich data sensors

• simple data sensors-

In this type the sensors can only exchange single data at every reporting time.

e.g. weight sensor, it only sense weight

• Rich data sensors:

This sensors can exchange multiple data at every reporting cycle.

e.g. engine sensors, it sense fuel, speed, carbon index etc. ^{in car}

5) Report Range:

- In this there are two sensors ^{e.g. smart car, microscope, camera} who can sense object from far

- another ~~is~~ need object too close to sense
 e.g. Fitness band

* Layer 2 - Communications N/W Layer: Access N/W Sublayer, Gateways and Backhaul sublayer, N/W Transport Sublayer, IOT N/W management Sublayer

→ Communication Network Layer:

- This is the layer that actually connects physical devices to the external network and make them relay smart and accesible over different network
- it use either wired or wireless ~~net~~ medium to connect physical devices.
- This layer have 4 sublayer

1) Access network sublayer

- this layer connect phy physical device to the network based on wireless technology e.g. Wifi, ZigBee, Bluetooth

2) Gateways and backhaul network sublayer

- it also connect physical device to the n/w e.g. cloud, Internet, Server
- sensor data will get transfer to external server for that gateways ~~ex~~ gives the interconnection
- IOT devices are functioning with different protocol like Wifi, bluetooth, ethernet, ZigBee etc. with this ~~devices~~ protocol IOT devices can communicate with IOT gateways

- IOT gateway can transfer this data to n/w i.e. cloud, Internet, server and the IOT gateway and N/W can communicate with Internet protocol i.e. TCP-IP, UDP etc.

- In router we just do data forwarding but by using IOT Gateway, IOT device transfer data to the server along with protocol transmission.

- IOT ^{Gateway} ~~device~~ can process data before transmission it forward only necessary data to the server

- IOT gateway forward the collected information through a longer range medium (called as backhaul)

3) Network transport ~~layer~~ sublayer:

This sublayer refers to the OSI model layer 3 (N/W layer) & 4 (transport layer), where TCP-IP and UDP protocol are used for sending and receiving the information

Thing could send the information via gateway and receive the control and management instructions via gateway.

e.g. temperature sensor in (fire alarm)

4) IOT n/w management sublayer

At this layer, additional networking protocols could be used to exchange data along with lower level protocol TCP-IP and UDP. some e.g. of higher level protocols could be CoAP (constrained Application protocol),

MQTT (Message Queuing Telemetry Transport)
it is restricted for low-bandwidth and high-latency IOT devices.

latency - delay in communication
Bandwidth - maximum data transfer rate

AMQP - Advanced Message Queuing protocol -
It passes business messages between applications or organizations.

3) Application and Analytics layer:

At this layer, there are several applications that process, analyse and visualise, control, manage and report the collected data.

- Analytics vs control application:

multiple applications can help increase the efficiency of an IOT network.

There are two types of application that work with IOT

① Analytics application:

this type of application collect data from multiple smart objects, process that data and display.

② Control application:

It control the action of the smart object
e.g. pressure sensor connected to the pump.

A control application ups the pump speed when connected sensor detect a drop in pressure.

Data Analytics vs N/W Analytics:
 Analytics means processing information to make sense of collected data.

There are two types of analytics

① Data Analytics:

The process of gathering data from various sensors analysing it and predicting something with it is data analytics.

② Network analytics:

It is the process where network data is collected and analyzed to improve the performance, reliability, visibility or security of network.

e.g. ope mines use wireless n/w to automatically pilot dump trucks. a loss of connectivity may result into the accident.

loss of ~~connel~~ connectivity means that data stop being fed to your data analytic platform

Types of Data analytics

- 1) Descriptive - it gathered data for knowledge
- 2) Diagnostic - it focuses on what happened and why
- 3) Predictive - it focuses on prediction & decision making
- 4) Prescriptive - it think about actions to be taken.